sometimes use products in ways that could harm them as well. For example, since organic fertilizers are difficult to obtain on a large scale in India, farmers often use farmyard manure, which may contain toxic chemicals and heavy metals. Certain plant sprays, such as Datura flower and leaf spray, have an element called atropine. If it is not applied in the right does, it can act on the nervous system of the consumer.

Unfortunately, how much and when to use it are not well-researched or regulated issues.

33. Based on the above passage, the following assumptions have been made:

1. Organic farming is inherently unsafe for both farmers and consumers.

2. Farmers and consumers need to be educated about eco-friendly food.

Which of the assumptions given above is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (d) : Neither statement 1 nor 2 is correct.

34. Which one of the following statements best reflects the most logical, rational and practical message conveyed by the author of the passage?

(a) In India, organic farming should not be promoted as a substitute for conventional farming.
(b) There are no safe organic alternatives to chemical fertilizers.
(c) In India, farmers need to be guided and helped to make their organic farming sustainable.
(d) The aim of organic farming should not be to generate huge profits as there is still no global market for its products.

Ans. (c) : Statement that best reflects the most logical, rational and practical message conveyed by the author of the passage is - "In India, farmers need to be guided and helped to make their organic farming sustainable".

Passage-3

Food consumption patterns have changed substantially in India over the past few decades. This has resulted in the disappearance of many nutritious foods such as millets. While food grain production has increased over five times since independence, it has not sufficiently addressed the issue of malnutrition. For long, the agriculture sector focused on increasing food production particularly staples, which led to lower production and consumption of indigenous traditional crops/grains, fruits and other vegetables, impacting food and nutrition security in the process. Further, intensive, monoculture agriculture practices can perpetuate the food and nutrition security problem by degrading the quality of land, water and food derived through them.

35. Based on the above passage, the following assumptions have been made:

A. To implement the Sustainable Development Goals and to achieve zero-hunger goal, monoculture agriculture practices are inevitable even if they do not address malnutrition.
B. Dependence on a few crops has negative consequences for human health and the ecosystem.
C. Government policies regarding food planning need to incorporate nutritional security.
D. For the present monoculture agriculture practices, farmers receive subsidies in various ways and government offers remunerative prices for grains and therefore they do not tend to consider crop diversity.

Which of the above assumptions are valid?

(a) 1, 2 and 4 only (b) 2 and 3 only
(c) 3 and 4 only (d) 1, 2, 3 and 4

Ans. (a) : Assumption 1, 2 and 4 are valid.

36. A box contains 14 black balls, 20 blue balls, 26 green balls, 28 yellow balls, 38 red balls and 54 white balls.

Consider the following statements:

1. The smallest number n such that any n balls drawn from the box randomly must contain one full group of at least one colour is 175.

2. The smallest number m such that any m balls draws from the box randomly must contain at least one ball of each colour is 167.

Which of the above statements is/are correct?

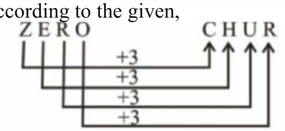
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (c) : <u>Statement 1</u>- Let the first 13 balls drawn from the box are black, next 19 balls are blue, next 25 are green, next 27 are yellow, next 37 are red, next 53 are white.
Now, in this process total balls drawn = 174.
If one more ball is drawn from the box, one colour group will be complete.
Hence, Statement 1 is correct.
<u>Statement 2</u>- Let the first 54 balls are white, next 38 are red, next 28 are yellow, next 26 are green, next 20 are blue. In this process, total number of balls drawn = 166.
In this group, no black ball is present. so the next ball drawn will be black and total number of balls will become 167.
Hence, statement 2 is also correct.

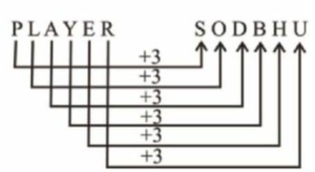
37. If 'ZERO' is written as 'CHUR', then how is 'PLAYER' written?

(a) SOCAGT (b) SODBGT
(c) SODBHT (d) SODBHU

Ans. (d) : According to the given,



So,



38. Consider the following statements:

1. A is older than B

2. C and D are of the same age

3. E is the youngest

4. F is younger than D

5. F is older than A

How many statements given above are required to determine the oldest person/persons?

(a) Only two (b) Only three
(c) Only four (d) All five

Ans. (d) : According to the given,
C = D > F > A > B > E
All the five statements are required to determine the oldest person.

39. Consider the following including the Question and the Staements :

There are 5 members A, B, C, D, E in a family

What is the relation of E to B?

Staement-1 : A and B are a married couple

Staement-2 : D is the father of C

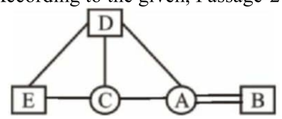
Staement-3 : E is D's son.

Staement-4 : A and C are sisters.

Which one of the following is correct in respect of the above Question and Statements?

(a) Statement-1, Statement-2 and Statement-3 are sufficient to answer the Question.
(b) Statement-1, Statement-3 and Statement-4 are sufficient to answer the Question.
(c) All four statements together are suffeicent to answer the Question.
(d) All four statements are not sufficient to answer the Question.

Ans. (c) : According to the given, Passage-2



Therefore, E is brother-in-law of B.
All four statements together are sufficient to answer the question.

40. Choose the group which is different from the others :

(a) 17, 37, 47, 97 (b) 31, 41, 53, 67
(c) 71, 73, 79, 83 (d) 83, 89, 91, 97

Ans. (d) : Options (a), (b) and (c) contain only prime numbers, while option (d) contains a composite number 91.

Direction for the following 3 (three) items :
Read the following three passages and answer the items that follow the passages. Your answers to these items should be based on the passages only.

Passage-1

Scientists studied the vernal window – transition period from winter to the growing season. They found that warmer winters with less snow resulted in a longer lag time between spring events and a more protracted vernal window. This change in the spring timetable has ecological, social and economic consequences – for agriculture, fisheries and tourism. As the ice melts earlier, the birds don't return, causing a delay, or lengthening in springtime ecological events.

41. With reference to the above passage, the following assumptions have been made :

1. Global warming is causing spring to come early and for longer durations.

2. Early spring and longer period of spring is not good for bird populations.

Which of the above assumptions is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (a) : According to the passage, "Global Warming is causing spring to come early and for longer durations" is correct.

Passage–2

A global analysis of nitrogen use efficiency –a measure of the amount of nitrogen a plant takes in to grow versus what is left behind as pollution – says that using to much fertilizers will lead to increased pollution of waterways and the air. Currently, the global average for nitrogen use efficiency is approximately 0.4, meaning 40 percent of the total nitrogen added to cropland goes into the harvested crop while 60 percent is lost to the environment, says a study. More than half of the world's population is nourished by food grown with fertilizers containing synthetic nitrogen, which is needed to produce high crop yields. Plants take the nitrogen they need to grow, and the excess is left in the ground, water and air. This results in significant emissions of nitrous oxide, a potent greenhouse and ozone depleting gas, and other forms of nitrogen pollution, including eutrophication of lakes and rivers and contamination of river water.

42. Which one of the following statements best reflects the most logical, rational and crucial message implied by the passage?

(a) An enhanced efficiency of use of nitrogen is imperative for both food production and environment.
(b) Production of synthetic nitrogen fertilizers cannot be stopped as it well adversely affect global food security.
(c) Alternatives to crops that require excess of nitrogen should be identified and cultivated.
(d) Conventional agriculture using synthetic fertilizers should be replaced with agroforestry, agro ecosystems and organic farming.

Ans. (a) : Statement that best reflects the most logical, rational and crucial message implied by the passage is- "An enhanced efficiency of use of nitrogen is imperative for both food production and environment".

Passage–3

Along with sustainable lifestyles, climate justice is regarded as a significant principle in environmental parlance. Both the principles have bearings on political and economic choices of the nation. So far, in our climate change summits or compacts, both the principles have eluded consensus among nations. Justice, in the judicial sense, is well defined. However, in the context of climate change, it has scientific as well as socio-political connotations. The crucial question in the next few years will be how resources, technologies and regulations are used to support the victims of climate change. Justice in climate is not confined to actions relating to mitigation, but includes the wider notion of support for adaptation to climate change and compensation for loss and damage.

43. Which one of the following statements best reflects the most logical, rational and crucial message conveyed by the passage?